

Clinical translation and engineering challenges of soft robotic cardiac sleeves for heart failure

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Soft robotic cardiac sleeves are an emerging class of non-blood-contact mechanical circulatory support devices designed to augment myocardial function while avoiding direct interaction with blood. Their development addresses important limitations of conventional heart failure therapies and offers a promising strategy for physiological cardiac assistance. This Review examines the pathophysiological basis for sleeve-assisted cardiac support, recent advances in soft robotic technologies, and the key engineering and translational challenges that must be overcome for clinical implementation. Finally, we propose a translational framework to guide the future development of clinically viable soft robotic cardiac sleeves.

Heart failure (HF) is a leading cause of morbidity and mortality worldwide, with an estimated global prevalence exceeding 64 million individuals, and it places a significant burden on healthcare systems through high rates of hospitalisation and associated costs^{1,2}. Although contemporary pharmacological therapies have improved survival, HF remains associated with a five-year mortality rate exceeding 50%, underscoring its progressive and life-limiting nature. Despite therapeutic advances, a significant proportion of patients progress to end-stage disease, characterised by irreversible myocardial damage and severely limited cardiac reserve³. Heart transplantation remains the definitive treatment for end-stage HF; however, its application is fundamentally constrained by the severe shortage of donor organs⁴.

Mechanical circulatory support (MCS) devices represent a principal alternative therapeutic strategy, providing mechanical support to the failing heart. However, only around 10–15% of patients qualify for device implantation owing to high surgical risk and complex prolonged management requirements^{5,6}. While contemporary MCS technologies can be life-saving, they are also associated with substantial complications, including thrombosis, infection, driveline failure, and the requirement for lifelong anticoagulation. Moreover, most contemporary ventricular assist devices (VADs) generate continuous, non-pulsatile blood flow that deviates from physiological haemodynamics, potentially contributing to additional complications^{7,8}. These

limitations have driven increasing interest in non-blood-contact approaches capable of augmenting cardiac function without direct interaction with circulating blood.

One emerging non-blood-contact strategy is the soft robotic cardiac sleeve, which provides external mechanical assistance by enveloping the heart⁹. Unlike conventional VADs that interact directly with blood flow, these systems augment cardiac function through synchronised external actuation of the myocardium^{9,10}. By avoiding direct blood contact, soft robotic sleeves may reduce clinical complications, while their compliant, modular architectures enable adaptation to patient-specific anatomy, pathology, and disease severity. Consequently, soft robotic sleeves have emerged as a promising adjunctive therapy for HF, alongside other advanced strategies such as genetically modified cardiac xenotransplantation, stem-cell-based therapies, and tissue-engineering approaches (Supplementary Table 1). Soft robotic cardiac sleeves can be viewed as a technology-enabled evolution of historical cardiomyoplasty concepts, replacing autologous skeletal muscle with engineered, electrically or pneumatically actuated artificial muscles to provide programmable, fatigue-resistant, and synchronised pulsatile epicardial assistance (Fig. 1a)^{11,12}.

Soft-robotic actuation strategies enable replication of the dynamic range, anisotropy, and spatiotemporal complexity of myocardial motion^{13,14}. Advances in bioelectronics and embedded sensing

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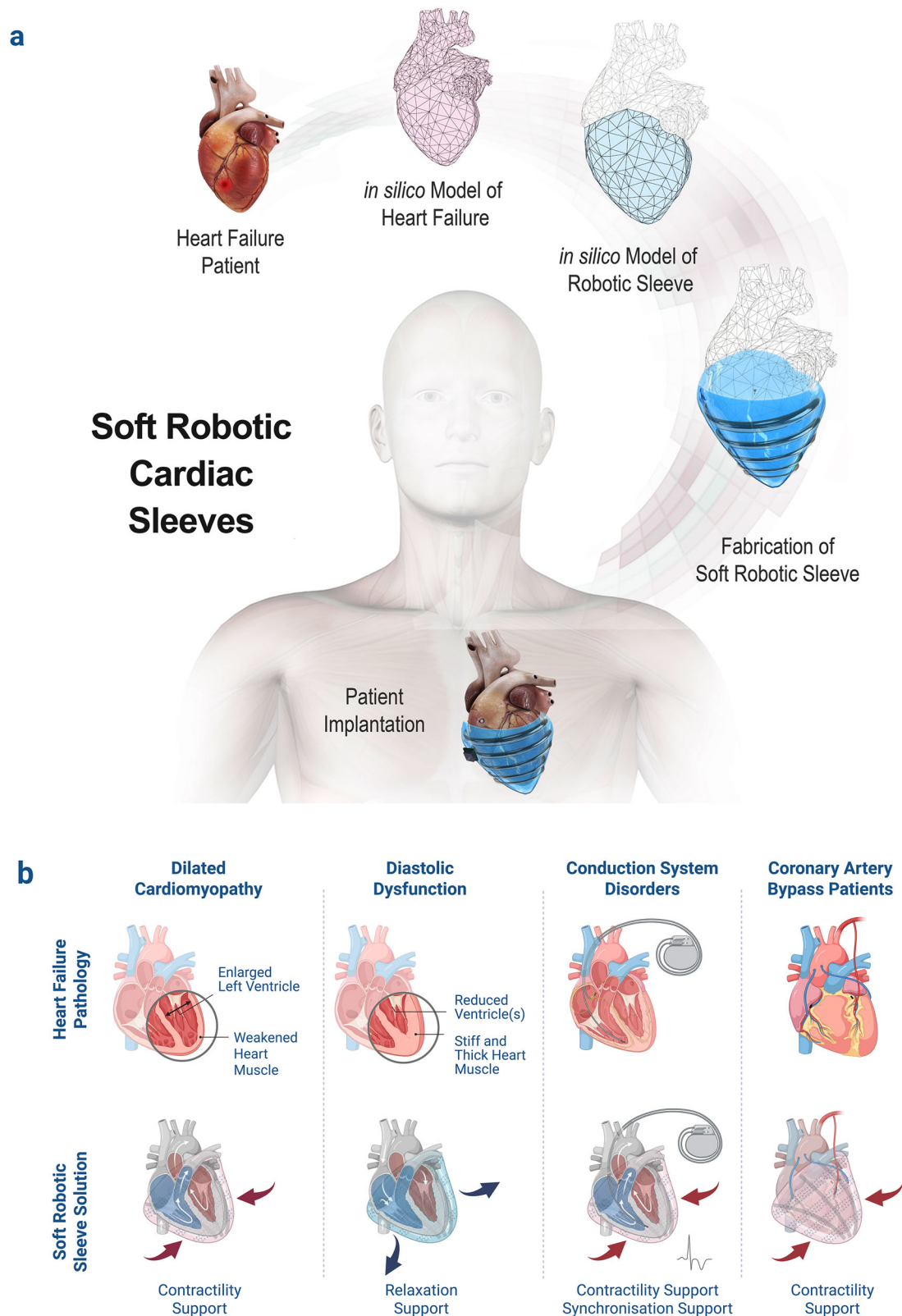


Fig. 1 | Translational pathway and potential clinical applications of soft robotic cardiac sleeves. **a** Conceptual development pipeline for soft robotic cardiac sleeves, progressing from patient-specific cardiac modelling and computational design to the fabrication of anatomically tailored devices and their implantation for physiological cardiac assistance. **b** Potential clinical applications of soft robotic cardiac sleeves. These devices provide non-blood-contact mechanical assistance through external compression, torsion and relaxation to support impaired myocardial function across multiple cardiac pathologies. In dilated cardiomyopathy, sleeve-

mediated compression may augment systolic function compromised by ventricular dilation and myocardial weakness. In diastolic dysfunction, sleeve actuation may assist ventricular relaxation and filling impaired by increased ventricular stiffness. Soft robotic sleeves may also support patients with cardiac conduction disorders by synchronising mechanical assistance with electrical pacing. In patients with severe ventricular dysfunction undergoing coronary artery bypass grafting, sleeve-based assistance may provide adjunct mechanical support following revascularisation. Created with BioRender.com (Nazari, H., 2026; <https://BioRender.com/rn98vea>).

facilitate closed-loop synchronisation of sleeve actuation with the cardiac cycle, as well as real-time monitoring of electrophysiological and arrhythmogenic activity¹⁵.

Although recent reviews have extensively addressed actuator technologies^{16,17}, materials development¹³, and mechanical design principles⁹, critical gaps remain in the discussion of in vivo performance, patient-specific implementation, and clinically viable translation pathways. In particular, the biological interface between soft robotic cardiac sleeves and the living myocardium (including vascular integration, immune response, and long-term tissue remodelling) remains underexplored within the predominantly engineering-focused literature, and tissue-engineering strategies capable of promoting durable integration and functional recovery have yet to be systematically incorporated. Despite growing interest in non-blood-contact cardiac assist technologies, the clinical translation of soft robotic cardiac sleeves is further constrained by interdependent challenges, including durable epicardial integration under chronic mechanical actuation, long-term biocompatibility, implantable and energy-efficient power and control systems, and reproducible surgical workflows compatible with diverse cardiac anatomies. Together, these limitations have restricted long-term animal studies and slowed progression toward clinical evaluation. Accordingly, this review integrates advances in soft robotics with emerging concepts in cardiac biology and regenerative medicine, presenting a comprehensive roadmap encompassing pathophysiological drivers of HF, engineering design considerations, preclinical validation strategies, and translational challenges to enable safe and practical medical application.

This review primarily focuses on active soft robotic cardiac sleeve systems that provide dynamic mechanical assistance through external epicardial actuation. Passive cardiac restraint devices and purely electronic cardiac support systems are discussed only where relevant to contextualise the evolution of mechanically assisted cardiac support technologies.

Pathophysiology of heart failure and role of soft robotic sleeves

HF is a complex clinical syndrome arising from insufficient cardiac pumping and/or filling capacity to meet systemic metabolic demands, resulting in systemic and pulmonary congestion¹⁸.

Clinically, HF is categorised into four primary phenotypes based on left ventricular ejection fraction (LVEF): HF with reduced ejection fraction (HFrEF; LVEF $\leq$ 40%), HF with preserved ejection fraction (HFpEF; LVEF $\geq$ 50%), HF with mildly reduced ejection fraction (HFmrEF; LVEF 41–49%), and HF with improved ejection fraction (HFimpEF)^{19,20}. While HFrEF is predominantly associated with systolic dysfunction, HFpEF arises from diastolic stiffening and impaired ventricular relaxation despite preserved contractility, and now accounts for ~50% of HF cases¹⁹. Pathophysiologically, advanced HF is characterised by reduced cardiac output, elevated ventricular filling pressures, impaired baroreflex sensitivity, and chronic activation of the renin-angiotensin-aldosterone and sympathetic nervous systems, which promote fibrosis, endothelial dysfunction, and metabolic derangements²¹.

Human cardiac transplantation remains the only curative option for selected patients with end-stage HF; however, its use is severely limited by donor organ scarcity and the requirement for lifelong immunosuppression²². Genetically modified porcine cardiac xenotransplantation has recently emerged as a potential strategy to address donor shortages, with early clinical attempts demonstrating short-term graft function but limited survival due to immune rejection and risks of zoonotic viral transmission²³. Device-based therapies, including VADs and total artificial hearts (TAHs), provide effective circulatory support but are associated with thrombotic, infectious, and driveline-related complications²⁴.

In contrast to conventional VADs, soft robotic sleeves envelop the myocardium and deliver controlled circumferential compression,

longitudinal shortening, and torsional deformation, closely replicating the physiological mechanics of systole and diastole⁹. In dilated cardiomyopathy, the ventricle is enlarged and weakened, with pronounced wall thinning (Fig. 1b), a characteristic of systolic failure in HFrEF, where the left ventricle fails to generate sufficient pressure to maintain systemic perfusion²⁵. Externally applied sleeve actuation may enhance myocardial contractility, augment stroke volume, and improve cardiac output, while restoring wall tension and mechanical synchrony.

In diastolic dysfunction, most commonly observed in HFpEF, ventricular stiffening impairs relaxation and elevates filling pressures (Fig. 1b)²⁶. Soft robotic sleeves capable of delivering timed, low-pressure recoil during early diastole could facilitate ventricular relaxation, improve filling, and alleviate pulmonary congestion.

In some cases, systolic dysfunction encompasses both acute and chronic ventricular failure states, characterised by overstretched, thin-walled ventricles with limited contractile reserve. While cardiac resynchronisation therapy improves electrical synchrony, it does not restore the complex physiological deformation patterns required for efficient ejection. Patient-specific soft robotic sleeves may enable regionally tailored mechanical actuation, reproducing native myocardial twisting and thereby enhancing systolic ejection while preserving diastolic filling²⁶.

Conduction system disorders, including heart block, atrial or ventricular fibrillation, and mechanical dyssynchrony, may similarly benefit from robotic sleeves functioning as mechanical synchronisers, contracting in coordination with intrinsic or paced cardiac rhythms (Fig. 1b)²⁷. Patients with advanced coronary artery disease or ischaemic cardiomyopathy frequently develop severe left ventricular dysfunction (ejection fraction $<$ 30%) and may undergo coronary artery bypass grafting (CABG). This high-risk patient population may also benefit from non-blood-contact mechanical assistance²⁷.

While the majority of soft robotic cardiac sleeve designs reported to date have primarily targeted left ventricular assistance, growing evidence from both clinical and preclinical studies highlights the critical role of right ventricular function, interventricular septal coupling, and pulmonary circulation in determining overall haemodynamic outcome. Isolated LV support (whether pump-based or externally applied) can unmask or exacerbate right ventricular failure, necessitating secondary RV support in a subset of patients. These considerations highlight the importance of biventricular mechanics when developing sleeve-based assist strategies. Recent studies on left and right atrioventricular coupling highlight the importance of ventricular filling dynamics, chamber compliance, pressure-volume relationships, and electromechanical synchrony in maintaining coordinated cardiac performance²⁸. Improved understanding of ventricular and atrioventricular coupling may contribute to the development of more physiologically adaptive cardiac sleeve systems. Next generation of soft robotic sleeves can benefit from architectures that enable differential, spatially resolved actuation across the left and right ventricles, or adaptive modulation of compression to preserve septal motion and pulmonary haemodynamic, particularly in patients with advanced or biventricular HF²⁸.

Broader biomechanical and translational considerations associated with direct cardiac compression technologies have also been comprehensively reviewed elsewhere²⁹. These analyses emphasise the importance of ventricular mechanics, haemodynamic coupling, and synchronised epicardial actuation in the development of physiologically responsive soft robotic cardiac assist systems²⁹.

Evolution of mechanical support strategies toward soft robotic sleeves

Mechanical circulatory support systems

Early supportive devices were developed to directly generate or sustain blood flow, giving rise to what is now broadly classified as MCS³⁰. MCS encompasses a range of technologies, including VADs, TAHs, and

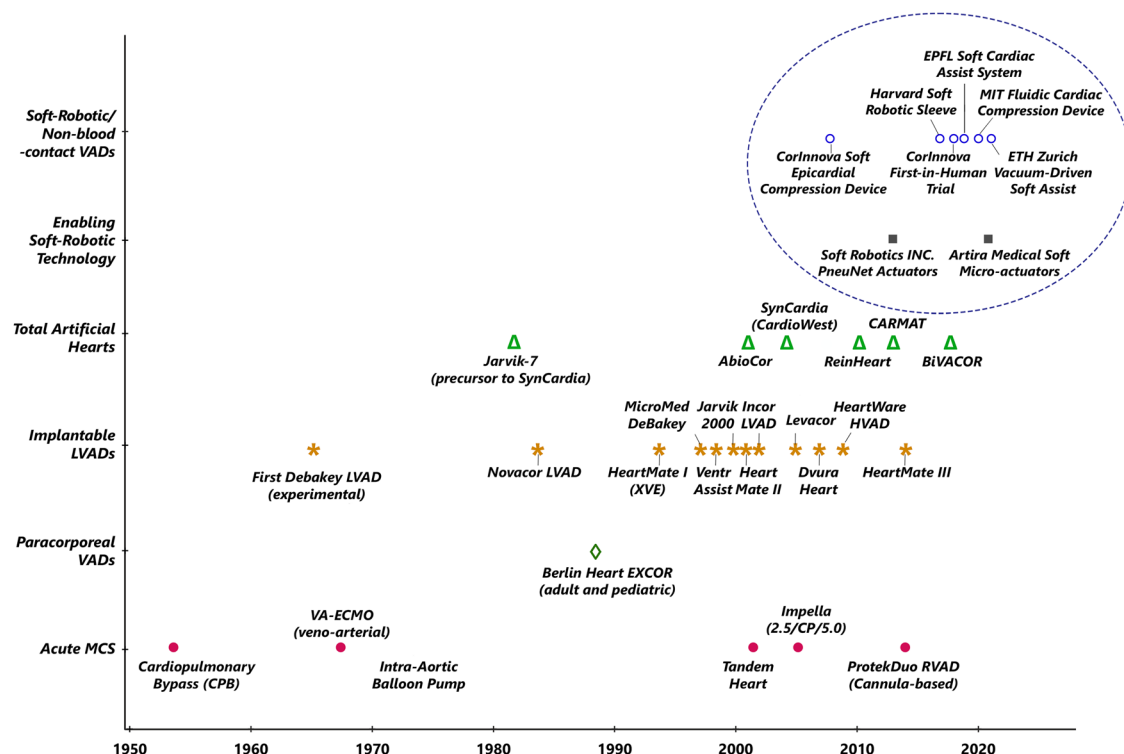


Fig. 2 | Evolution of mechanical circulatory support technologies. Timeline of major mechanical circulatory support (MCS) systems from 1953 to 2025, illustrating the progression from acute support devices and paracorporeal ventricular assist devices (VADs) to implantable VADs and total artificial hearts (TAHs). The figure also highlights enabling technologies that have contributed to the development of

soft robotic cardiac assistance, culminating in recent non-blood-contact soft robotic cardiac assist systems. Symbols denote acute MCS devices (●), paracorporeal systems (◊), implantable VADs (*), total artificial hearts (△), enabling soft robotic technologies (■), and soft robotic non-blood-contact VADs (◐).

extracorporeal systems, designed to augment or replace cardiac pumping function in patients with severe HF (Fig. 2). Over several decades, these technologies have evolved from bulky extracorporeal pumps introduced in the 1960s to compact hard robotic devices, emerging soft robotic sleeves, and fully implantable artificial heart systems (Fig. 3a)^{31,32}. Commercial VAD platforms, including the HeartMate and HeartWare HVAD series, are now established as bridge-to-transplant or destination-therapy options; however, their direct interaction with circulating blood continues to constrain long-term outcomes³³.

Hard robotic assist devices, such as axial- and centrifugal-flow VADs, represent the engineering apex of pump-based circulatory assistance, offering compact form factors, magnetic levitation, and multi-year operational reliability³⁴. Nevertheless, their rigid architectures and high-shear flow environments differ fundamentally from native myocardial mechanics. In parallel, the development of TAHs, including the SynCardia™, CARMAT™, and AbioCor™ systems, pursued complete ventricular replacement but has been limited by device size, biocompatibility constraints, and long-term integration challenges³⁵. These limitations prompted renewed interest in non-blood-contact assist strategies capable of acting externally to the cardiac chambers. Early concepts, including pneumatic compression cuffs and experimental epicardial sleeves, laid the foundation for soft robotic approaches that support myocardial motion through gentle epicardial compression rather than intravascular pumping¹⁰.

Soft robotic cardiac sleeves

Soft robotic cardiac sleeves represent the latest class of mechanical cardiac assist devices, designed to replicate physiological myocardial contraction through external epicardial actuation. In contrast to conventional VADs that actively propel blood via internal pumps, soft

robotic sleeves envelop the epicardial surface and externally augment ventricular contraction, thereby reducing the risks of thrombosis, haemolysis, and infection^{10,36}. Broadly, soft robotic cardiac sleeves can be classified into pneumatic systems, thermally driven systems, including shape memory alloy (SMA) and twisted and coiled polymer (TCP) actuators, and electroactive polymer (EAP) systems¹⁶. Although these systems emphasise compliant biointerfaces and deformable actuation, many emerging platforms incorporate hybrid hard/soft architectures integrating sensing modules, control electronics, anchoring structures, and power-delivery components to achieve synchronised and reliable cardiac assistance. (Fig. 3b) Related soft robotic artificial muscle-based cardiac assist systems have also been reported, including ventricular simulators capable of reproducing myocardial biomechanics; however, these platforms primarily function as heart replacement or simulation devices rather than conformal epicardial sleeves, and are therefore not classified within sleeve-based approaches discussed here³⁷.

Elements of soft robotic cardiac sleeves

As the structural backbone of the sleeve, the soft base layer conforms intimately to the epicardial surface and distributes actuation forces uniformly across the ventricular wall (Fig. 3b)¹⁰. To minimise mechanical mismatch, the substrate must approximate myocardial compliance (elastic modulus $E \approx 20\text{--}50\text{ KPa}$), accommodate 10–15% physiological strain, and retain sufficient tensile strength for continuous cyclic loading³⁸.

Actuators are the active contractile elements that convert external energy into mechanical motion, enabling reproduction of myocardial contraction¹⁷. Among the various strategies explored, pneumatic sleeves represent the most mature preclinical and translational platform, forming the basis of many contemporary soft robotic cardiac assist systems¹⁰.

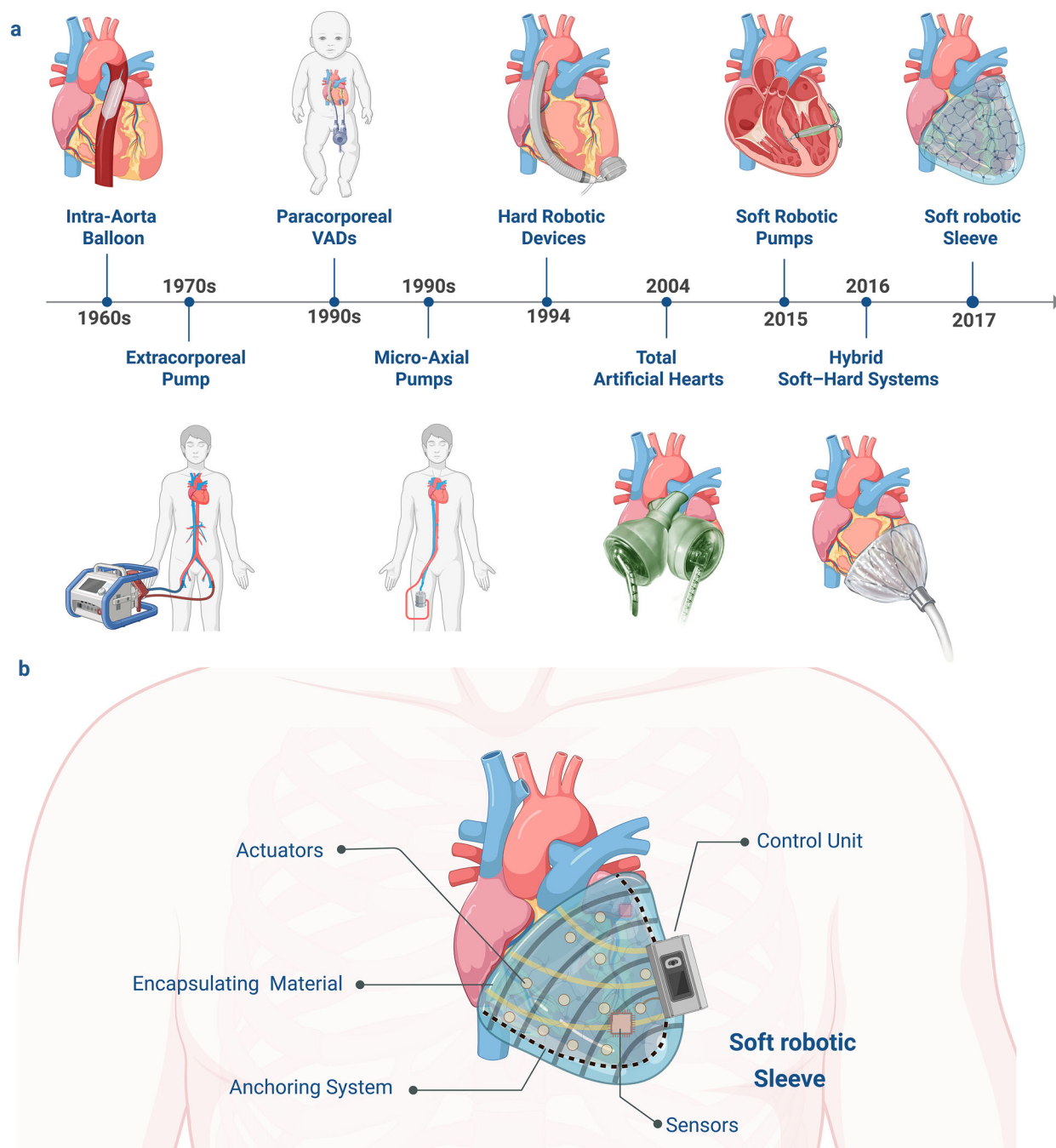


Fig. 3 | Evolution and core components of soft robotic cardiac sleeve systems. **a** Timeline illustrating the progression of mechanical circulatory support technologies, from extracorporeal pumps and intra-aortic balloon pumps to implantable ventricular assist devices, total artificial hearts and hybrid systems, culminating in the development of non-blood-contact soft robotic cardiac sleeves. **b** Conceptual

architecture of a soft robotic cardiac sleeve, highlighting key components enabling closed-loop cardiac assistance, including actuators, encapsulating materials, anchoring systems, integrated sensors, control unit, and drive line. Created in BioRender. Nazari, H. (<https://BioRender.com/rn98vea>).

SMA-based sleeves exploit the thermally induced phase transformation of nickel-titanium (NiTi) alloys, which contract upon electrical heating and relax during cooling^{39,40}. These systems enable compact, silent actuation without external pumps, though thermal inertia and energy efficiency remain limiting factors⁴¹. TCP actuators employ electrically heated polymer fibres, such as nylon or polyethylene terephthalate, twisted into muscle-like bundles capable of producing large linear strain⁴². EAP sleeves utilise electric-field-induced deformation of dielectric or ionic polymer films, producing rapid, electronically controlled actuation⁴³. Pneumatic systems remain

the most clinically advanced, whereas SMA, TCP, and EAP technologies continue to evolve. Hybrid actuation strategies aim to combine the force capacity of pneumatic systems with the precision of electro-active or thermal actuation; however, system complexity and immature integration presently limit translational readiness⁴⁴.

Secure attachment of the sleeve to the epicardial surface is essential for mechanical efficiency, positional stability, and long-term safety. Suction-based anchoring systems utilise negative-pressure zones to maintain conformal contact without tissue penetration, reducing trauma and facilitating device removal^{9,45}. Hydrogel and

silicone-based adhesives enhance adhesion while minimising micro-motion and fibrotic encapsulation¹⁰. Supplementary structural supports, including sutures or polymeric rings positioned at the ventricular base or pericardial margins, provide additional stability for prolonged operation⁴⁶. Hybrid anchoring strategies, combining suction and limited suturing, have demonstrated improved fixation and durability in large-animal models without compromising epicardial perfusion⁴⁷.

Embedded sensors enable closed-loop control, synchronising sleeve actuation with the native cardiac cycle while providing continuous performance feedback. Epicardial electrocardiogram (ECG) electrodes detect R-wave timing to trigger systolic compression and are typically fabricated from platinum-iridium or gold-coated silicone integrated within flexible circuits¹⁰. Mechanical sensors, including piezoresistive, capacitive, and strain-based devices (often using carbon-black or liquid-metal composites) monitor curvature, wall strain, and actuator displacement. Pressure sensors, commonly microelectromechanical systems (MEMS)-based capacitive diaphragms, record internal chamber pressures to support safety monitoring and adaptive control⁴⁸.

Encapsulation strategies combining medical-grade silicone with parylene-C or polyurethane barrier layers provide effective multilayer moisture protection⁴⁹. Controllers coordinate actuation sequences and synchronise sleeve contraction with ventricular systole, most commonly using ECG-gated control algorithms⁵⁰.

The driveline provides the interface between internal components and external control hardware. Most current prototypes employ percutaneous drivelines, analogous to those used in VAD systems, to transmit pneumatic pressure and sensor signals⁵¹. However, the risk of chronic infection has driven the development of fully implantable, wireless interfaces utilising inductive power transfer and telemetry.

Transcutaneous energy transfer (TET) systems have emerged as a promising strategy for eliminating percutaneous drivelines and reducing the associated risks of chronic infection in implantable cardiac assist devices^{52,53}. These systems typically utilise inductive coupling between external and implanted coils to wirelessly transmit electrical power across the skin, enabling fully implantable platforms without permanent transcutaneous connections⁵⁴. In addition to powering pneumatic or electromechanical actuation systems, TET platforms may also support bidirectional telemetry for physiological monitoring, sensor feedback, and closed-loop device control⁵⁴. Similar concepts have been explored in VADs and other implantable bioelectronic systems to improve patient mobility and reduce infection-related complications. Despite their significant translational potential, challenges including energy-transfer efficiency, tissue heating, coil misalignment, battery longevity, and long-term system reliability remain important considerations for chronic implantation and clinical translation^{55,56}.

Characterisation of soft robotic sleeves, current methods

Development of soft robotic cardiac sleeves typically involves computational design, material optimisation, benchtop validation, and ex vivo and in vivo assessment⁵⁷. (Fig. 4). The characterisation process is often disease-oriented, beginning with computational modelling and finite element analysis (FEA) to predict stress distribution, actuator strain, and global sleeve deformation under simulated cardiac loading conditions⁵⁸. Anatomical reconstruction from patient-derived computed tomography (CT) or magnetic resonance imaging (MRI) datasets facilitates the generation of personalised ventricular geometries, improving anatomical conformity and minimising shear stress at the epicardial interface (Fig. 4)^{51,59}. Following the computational stage, material characterisation is performed to determine the mechanical properties, fatigue resistance, and durability of candidate elastomers, most commonly medical-grade silicones⁶⁰ and TPU^{10,45,61}.

Once fabricated (most commonly via three-dimensional (3D) printing, soft lithography, or moulding techniques) the sleeve undergoes geometric and conformational validation using 3D-printed or silicone ventricular phantoms. The ex vivo characterisation phase subsequently evaluates sleeve performance in a physiologically relevant environment⁶². The device is mounted onto a compliant ventricular phantom integrated within a closed-loop hydrodynamic circulation system that replicates cardiac filling and ejection. Key performance metrics (including generated pressure, stroke volume, and volumetric compression ratio) are quantified using pressure transducers, flow sensors, and optical motion-tracking systems.

The long-term objective of soft robotic cardiac sleeves is to achieve safe, durable, and physiologically synchronised function in the living heart. Although in vitro and ex vivo testing provide essential insight into mechanical performance and fluid dynamics, in vivo evaluation represents the critical translational step linking engineering function with biological response^{46,63}. Initial in vivo evaluation focuses on histological and biological analyses, examining tissue integration, fibrotic encapsulation, inflammatory cell infiltration, and vascularisation at the sleeve-epicardium interface. Effective integration minimises shear stress and motion-induced irritation, while the development of microvascular networks (vasculogenesis) within the interface layer may enhance nutrient exchange and mitigate chronic inflammation. Conversely, excessive fibrosis, immune activation, or local ischaemia can compromise both mechanical performance and myocardial health.

Alongside biological assessment, functional performance is evaluated using a combination of invasive and non-invasive cardiological diagnostics. Echocardiography and ultrasound imaging enable real-time quantification of stroke volume, ejection fraction, and left ventricular end-diastolic volume (LVEDV), reflecting the sleeve's capacity to restore contractility and diastolic filling^{10,51}. Pressure probes inserted into the ventricular and aortic chambers enable measurement of transaortic pressure gradients and ventricular loading conditions. Thermal safety is assessed using infrared thermography or implanted temperature sensors, ensuring that actuation-induced heating remains within clinically acceptable limits for chronic operation (Fig. 4).

Current progress in preclinical soft robotic cardiac sleeves

Over the past decade, research in soft robotic cardiac assistance has progressed from benchtop feasibility studies to increasingly sophisticated preclinical platforms capable of reproducing the mechanical and haemodynamic signatures of HF. These studies demonstrate a broader transition from proof-of-concept actuation devices towards biologically informed, physiologically adaptive, and translationally oriented cardiac assist systems.

The first major step in this direction was reported by Roche and colleagues at the Wyss Institute and Boston Children's Hospital, who developed a direct cardiac compression (DCC) device designed to replicate native myocardial contraction using soft robotic actuation⁶¹. Multiple materials were evaluated for the internal bladders of pneumatic artificial muscles, including silicone, polyethylene terephthalate (PET), nylon, and TPU, to identify optimal mechanical compliance and durability. This study established the feasibility of conformal epicardial soft robotic sleeves capable of generating physiologically relevant volumetric outputs while closely matching cardiac geometry⁶¹.

Building on this foundation, the translation of this concept into fully implantable soft robotic sleeves tested in vivo in large-animal models represented a major milestone in the field (Fig. 5a)¹⁰. Thin silicone laminate sleeves integrating circumferential and helical TPU pneumatic artificial muscles (PAMs) enabled simultaneous compressive and torsional assistance that more closely replicated native myocardial fibre orientation and ventricular twist mechanics. Acute in vivo

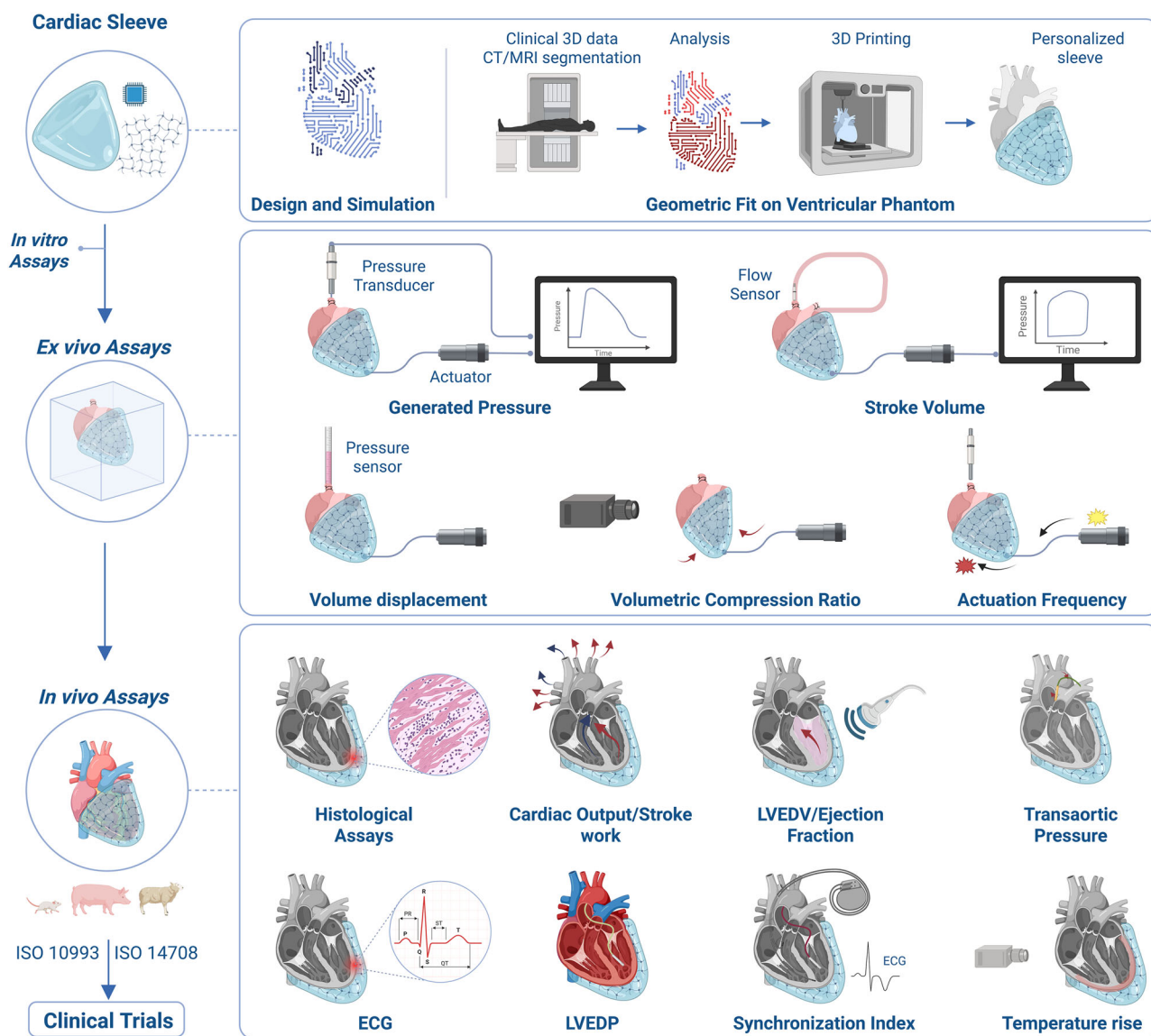


Fig. 4 | Workflow for the design, fabrication and multi-level evaluation of soft robotic cardiac sleeves. Conceptual framework illustrating the translational development pathway of soft robotic cardiac sleeves. Patient-specific devices are generated through in silico modelling, clinical CT or MRI image segmentation, geometric optimisation and additive manufacturing to produce anatomically tailored sleeves. Ex vivo testing involves mounting the sleeve on ventricular phantoms within hydrodynamic systems to assess pressure generation, stroke volume,

volumetric compression and actuation frequency. In vivo evaluation (conceptual stage) includes assessment in large-animal models, examining haemodynamic performance and safety parameters such as cardiac output, stroke work, left ventricular end-diastolic pressure (LVEDP), ejection fraction, transaortic pressure, ECG synchronisation, thermal effects and histological integration. Created in BioRender. Nazari, H. (<https://BioRender.com/rn98vea>).

studies in Yorkshire swine demonstrated substantial restoration of cardiac output and ventricular performance under pharmacologically induced HF conditions¹⁰. At the same time, these investigations identified several critical translational challenges, including inflammatory responses at fixation interfaces, chronic biointegration, and long-term mechanical durability under cyclic loading¹⁰.

A more conceptual advance was introduced by Horvath and colleagues, who identified stable epicardial integration as a fundamental determinant of long-term device performance (Fig. 5b)⁴⁶. Their work highlighted that effective haemodynamic support depends not only on actuator force generation, but also on durable mechanical and biological coupling between the device and the epicardial surface. Surgical mesh interfaces promoted vascularisation and tissue integration, whereas silicone-based interfaces resulted in encapsulation and comparatively poor adhesion⁴⁶. These findings reinforced the

importance of tissue-interface engineering as a central translational consideration for chronic soft robotic cardiac assistance.

Alternative assistance strategies have also emerged that move beyond complete ventricular encasement. Payne and colleagues developed an implantable extracardiac soft robotic system incorporating McKibben pneumatic artificial muscles capable of providing both systolic compression and diastolic recoil without circumferential enclosure (Fig. 5c)⁶⁴. Importantly, this platform introduced physiologically responsive actuation triggered by intraventricular pressure changes rather than relying exclusively on ECG synchronisation, thereby advancing the concept of adaptive closed-loop cardiac assistance. Acute in vivo studies demonstrated marked improvements in cardiac output and aortic flow under optimised coupling conditions⁶⁴. However, the absence of chronic implantation, durability testing, and long-term biocompatibility assessment limited conclusions regarding

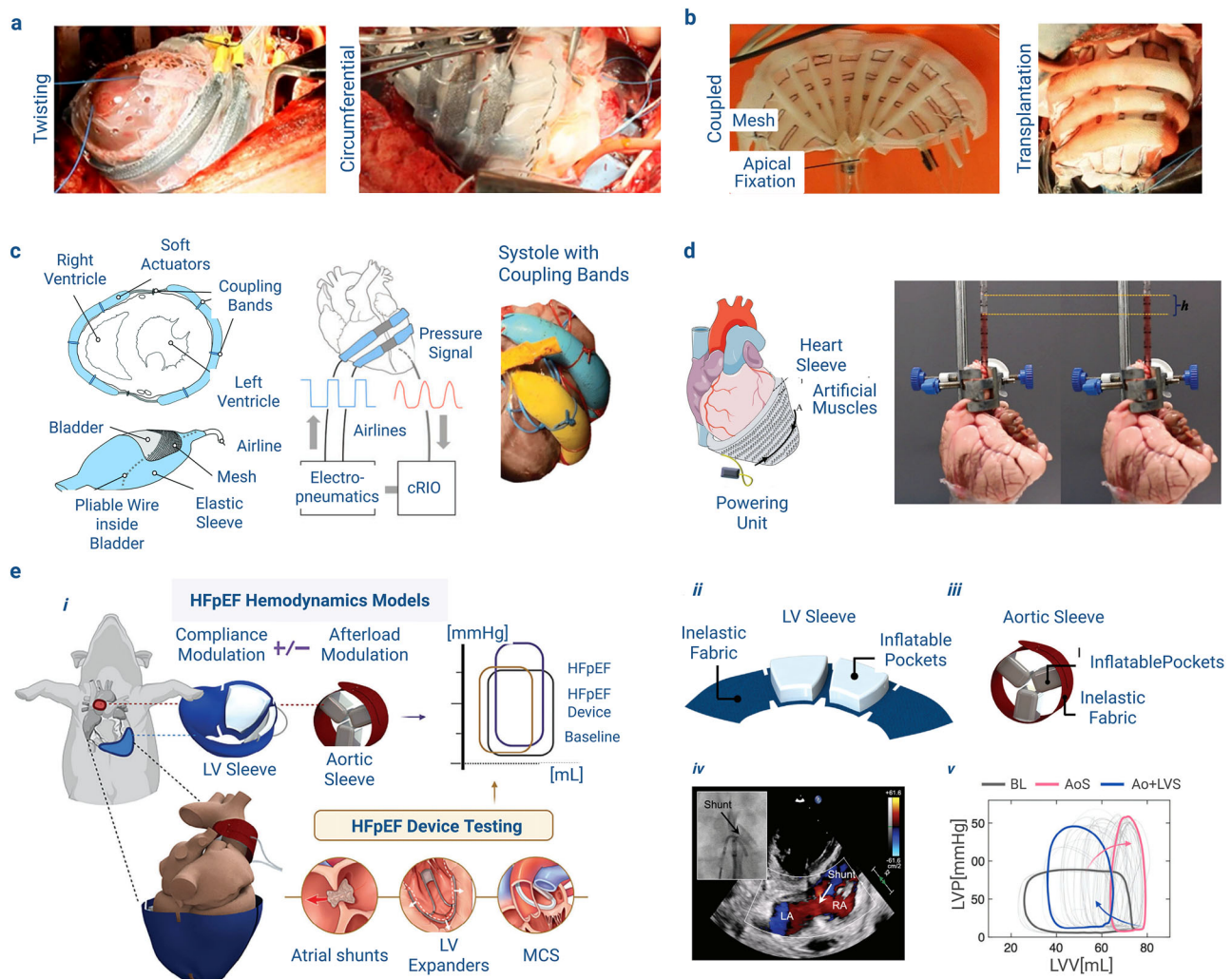


Fig. 5 | Representative soft robotic cardiac sleeve prototypes and translational studies. **a** Porcine demonstrations of electrocardiogram (ECG)-triggered twisting and circumferential actuation, illustrating synchronised ventricular assistance using soft robotic sleeves. A hydrogel-mediated coupling layer enables conformal, low-friction contact between the device and epicardial surface during in vivo assistance. Adapted from ref. 10. **b** Implantation of a “coupled” adhesive fixation system in porcine models, enabling direct coupling of the sleeve to the myocardium and restoration of cardiac output. Adapted from ref. 46. **c** Implantable extracardiac sleeve integrating soft actuators, coupling bands and an elastic bladder for synchronised ventricular assistance, including device layout and control

architecture. Adapted from ref. 64. **d** Artificial muscle-based sleeves using silver- and silicone-coated actuators tested ex vivo in a lamb heart model, demonstrating blood-free coupling and contractile assistance. Adapted from ref. 67. **e** Translational large-animal testing platforms for heart failure (HF) with preserved ejection fraction (HFpEF). Tunable porcine haemodynamic models incorporate left ventricular (LV) and aortic sleeves with atrial shunts to modulate compliance and afterload. Structural sleeve designs (i–iii) and representative pressure-volume loop analyses (iv–v) illustrate device-mediated haemodynamic modulation. Adapted from ref. 45.

sustained clinical applicability. Despite these advances, current sleeve actuation strategies still do not fully replicate the coordinated myocardial pressure generation and deformation patterns of the native myocardium. In particular, synchronisation between externally applied sleeve actuation and intrinsic ventricular electromechanical activity remains incomplete, remaining a major unresolved engineering challenge for cardiac sleeve systems (Fig. 5c).

Chronic epicardial compression may also introduce important physiological risks that remain insufficiently characterised, including impaired coronary perfusion, altered ventricular compliance, arrhythmogenic electromechanical disturbances, or constrictive remodelling over prolonged implantation periods⁶⁵. In addition to systolic augmentation, several platforms have explored active diastolic assistance through controlled recoil or decompression mechanisms intended to facilitate ventricular filling and reduce diastolic wall stress, although the long-term physiological consequences of such strategies remain incompletely understood.

Additional approaches have explored alternative actuation and power-transfer mechanisms. Han and colleagues proposed a muscle-powered biventricular assist device (Bi-VAD) coupling a latissimus dorsi-driven muscle energy converter (MEC) to a DCC sleeve⁶⁶. This biohybrid strategy illustrated the potential for biologically integrated energy-transfer systems capable of generating physiologically relevant ventricular compression while reducing dependence on entirely externalised power systems⁶⁶.

In parallel, Foroughi and colleagues investigated electrothermal artificial muscles capable of generating pressures approaching physiological left ventricular systolic pressure ranges (Fig. 5d)⁶⁷. Silver-coated nylon actuators embedded within silicone elastomer matrices generated considerable contractile pressures and ventricular-like volumetric displacement in ex vivo models. These studies highlighted the promise of compact electrothermal actuation systems for future soft robotic cardiac devices, while simultaneously underscoring unresolved challenges related to

BOX 1**Clinical decision-making for non-blood-contact cardiac assistance**

Clinical benefit from non-blood-contact cardiac assist technologies depends on aligning patient phenotype, therapeutic intent and timing of intervention.

Key considerations:

- What is the primary therapeutic goal? Stabilisation, bridge to recovery, bridge to transplantation, or destination support
- Which HF phenotype is most likely to benefit? Predominantly systolic dysfunction, preserved diastolic reserve, biventricular involvement
- When should epicardial assistance be initiated relative to disease progression or acute decompensation?
- How should treatment response be assessed? Haemodynamics, symptoms, hospitalisation burden, functional capacity
- What criteria should guide escalation, continuation or withdrawal of external mechanical assistance?

thermal safety, inflammatory response, and long-term tissue compatibility⁶⁷.

More recently, Rocchi and Rosalia expanded soft robotic systems beyond therapeutic assistance towards disease-modelling and device-testing platforms. A hybrid *in silico-in vitro* framework was introduced using a patient-specific echogenic soft robotic left ventricle reconstructed from cardiac magnetic resonance (CMR) data and integrated into a closed-loop cardiovascular simulator⁵¹. This system reproduced haemodynamics of HFpEF, dilated cardiomyopathy, and normal physiology with high fidelity, while enabling direct four-dimensional echocardiographic validation. Notably, the platform demonstrated predictive capacity by evaluating transcatheter aortic valve replacement (TAVR) devices, accurately reflecting improvements in effective orifice area, stroke volume, flow velocity, and reduction of paravalvular leak⁶⁸. Such developments demonstrate how soft robotic cardiovascular technologies may increasingly function not only as assist devices, but also as predictive translational platforms for procedural planning, device optimisation, and mechanistic cardiovascular research.

Subsequent work by Rosalia and colleagues advanced translational modelling by developing tunable soft robotic sleeves to reproduce HFpEF haemodynamics *in vivo* in Yorkshire swine (Fig. 5e)⁴⁵. Acute modulation of ventricular compliance and aortic loading successfully created characteristic HFpEF haemodynamics and enabled evaluation of interatrial shunting strategies under physiologically relevant conditions⁴⁵. Despite these advances, experiments were limited to acute timeframes, leaving chronic remodelling, durability, and long-term biointegration unaddressed.

Progress in biohybrid cardiovascular systems has expanded opportunities for physiologically relevant disease modelling and pre-clinical device evaluation. In addition to soft robotic cardiac assist platforms, organosynthetic and biohybrid cardiac models have demonstrated the integration of engineered actuation systems with biologically relevant cardiac architectures to reproduce complex myocardial motion and intracardiac biomechanics under controlled *in vitro* conditions. These systems incorporate biomimetic ventricular geometries, myocardial fibre alignment, diffusion tensor imaging-guided architectures, and biohybrid actuation strategies to emulate native cardiac deformation and haemodynamic behaviour with enhanced physiological relevance^{28,69,70}. Notably, Park et al. developed an organosynthetic dynamic heart model guided by cardiac diffusion tensor imaging that preserved anatomically realistic intracardiac structures while reproducing physiologically representative ventricular motion for procedural simulation and intracardiac device testing⁶⁹. Similarly, recent robotic right ventricle and biohybrid actuator platforms have demonstrated the ability to simulate right ventricular function under physiological and pathological conditions while integrating biologically derived muscular systems with robotic control architectures^{28,70}.

Emerging efforts toward soft robotic and hybrid heart systems, including recent developments from the Hybrid Heart initiative,

highlight the increasing integration of soft robotics, tissue engineering, and biomaterial-based cardiovascular support technologies⁷¹. A recent proof-of-concept hybrid heart combined pneumatically driven soft robotic actuation with supramolecular polymeric materials to improve biocompatibility while reproducing adaptive cardiac pumping behaviour⁷¹. Although primarily developed as a bioartificial heart replacement platform, this work demonstrates the broader translational potential of biologically integrated soft robotic cardiovascular systems and the continued evolution of adaptive, bioinspired cardiac support technologies⁷¹.

Current preclinical studies suggest that soft robotic cardiac sleeves have progressed from proof-of-concept mechanical systems to biologically informed platforms capable of restoring physiological haemodynamics, interrogating disease-specific pathophysiology, and enabling device testing in large-animal models. At the same time, these investigations consistently reveal unresolved limitations, including challenges in achieving durable epicardial biointegration, maintaining long-term mechanical stability under cyclic loading, ensuring inflammatory and thermal safety, and managing control and system complexity. Notably, most studies remain confined to acute or short-duration experiments, with a lack of chronic large-animal data underscoring the translational gap between short-term feasibility and sustained clinical application (Box 1). These constraints define the context for the emerging technological trends, design principles, and translational frameworks discussed in the following sections. Future translational studies will require chronic large-animal evaluation of long-term device durability, inflammatory response, ventricular reverse remodelling, and survival outcomes under sustained physiological loading conditions.

Emerging trends and future directions**Integrating tissue interface into soft robotic cardiac sleeves**

Despite substantial progress in soft robotic cardiac sleeve design, significant gaps remain in long-term *in vivo* validation, histological assessment, and durable biological integration. Current systems primarily deliver short-term mechanical augmentation, whereas future development increasingly points toward biohybrid strategies capable of supporting sustained functional recovery through biologically mediated mechanisms.

Recent first-in-human experience underscores the importance of biologically informed epicardial interfaces in non-blood-contact cardiac assist systems. In a clinical feasibility study, Wildhirt et al. demonstrated a biventricular, pulsatile extravascular MCS platform based on a conformal cardiac sleeve incorporating epicardial electrodes and pneumatic cushions⁷². Real-time epicardial ECG sensing and adjustable deflation timing relative to aortic valve closure enabled optimisation of ventricular filling while avoiding electromechanical asynchrony⁷². The technology initially developed by Wildhirt and colleagues has subsequently progressed toward clinical translation through the ReBeat system developed by Adjucor GmbH, representing

one of the first direct cardiac compression platforms to advance to first-in-human clinical investigation for advanced HF management⁷³. Unlike conventional blood-contacting VADs, the ReBeat system provides synchronised pulsatile epicardial compression using inflatable cushions and epicardial ECG-guided mechanical synchronisation while avoiding direct blood contact⁷³. Recent first-in-human clinical investigations demonstrated the translational feasibility of this biventricular mechanical circulatory support strategy for patients with advanced HF requiring durable ventricular assist support⁷³. Earlier perspectives from Criscione highlighted the emerging potential of soft robotic direct cardiac compression systems capable of restoring cardiac mechanics through biomimetic epicardial actuation while avoiding direct blood contact⁷⁴.

While existing pneumatic, hydraulic, and electroactive sleeves effectively reproduce aspects of cardiac motion, their interaction with the myocardium remains largely mechanical, underscoring the importance of the epicardial interface as a biological determinant of long-term device performance (Supplementary Note 1). Conceptual frameworks proposed by Duffy and Feinberg provide a bridge between soft robotics and regenerative biohybrid systems by emphasising the recreation of cellular microenvironments through controlled scaffold stiffness, extracellular matrix composition, and mechanical or electrical conditioning⁷⁵. Translating such principles to cardiac applications suggests that future sleeves may require biomimetic architectures capable of promoting synchronised contraction, vascularisation, and long-term myocardial remodelling.

Advances in micro- and nanofabrication (including electrospinning, three-dimensional bioprinting, and decellularised tissue processing) enable the construction of layered and fibre-aligned architectures that support compliance, metabolic exchange, and vascular infiltration (Fig. 6a)⁷⁶. Recent anisotropic elastin-like hydrogels reinforced with silk fibroin fibres demonstrate how fibre orientation combined with stimuli-responsive biopolymers can generate directional actuation and myocardial-like mechanical anisotropy⁷⁷. Electrically conductive components (including carbon-based nanomaterials, metallic nanoparticles, two-dimensional nanosheets, and electroactive polymers) represent an additional emerging direction to improve electromechanical coupling at the tissue-device interface^{78,79}. However, their long-term biostability, potential cytotoxicity, and mechanical compatibility with dynamic cardiac tissue remain incompletely characterised. Parallel advances in soft bioelectronics, including graphene- and gold-nanowire-based composites, demonstrate enhanced conductivity and reduced interfacial impedance without sacrificing compliance, but chronic *in vivo* evaluation remains limited^{80,81}. In parallel, textile-integrated fabrication strategies are also emerging as promising approaches in soft robotic systems. Knitted, braided, and woven fibre-based architectures can provide enhanced mechanical anisotropy, structural durability, and deformation control while maintaining flexibility and conformability to dynamic cardiac surfaces⁸². Such textile-reinforced systems may additionally improve fatigue resistance and support more scalable manufacturing pathways for functional implantable soft robotic cardiac sleeves⁸³. Developments in textile-based soft actuators and sensing platforms highlight the translational potential of integrating smart fibre networks within bioinspired cardiac assist devices⁸⁴.

Beyond structural and electrical integration, biofunctionalisation strategies are increasingly explored. Controlled local delivery of growth factors, anti-fibrotic agents, antibiotics, microRNAs, or extracellular vesicles may promote angiogenesis, modulate inflammation, and guide matrix remodelling at the epicardial interface^{85,86}. Incorporation of living cells (such as cardiomyocyte sheets, endothelial layers, or stem-cell-derived progenitors) offers the potential for dynamic biological coupling and paracrine signalling, but substantially increases fabrication complexity, cost, and regulatory burden^{87,88}. Challenges including cell viability during manufacture and

implantation, immune compatibility, vascularisation, and electrical synchrony remain unresolved.

Experience from cardiac tissue engineering indicates that cell-based myocardial constructs have achieved only partial functional restoration in preclinical and early clinical settings, often limited by poor retention, insufficient vascularisation, and electrical mismatch⁸⁹. These observations suggest that near-term translational strategies may favour acellular or transiently bioactive approaches rather than fully cellularised sleeves. Collectively, these trends indicate a shift toward increasingly complex biohybrid systems, while simultaneously exposing challenges related to scalability, reproducibility, and long-term integration that remain to be addressed (Supplementary Note 2).

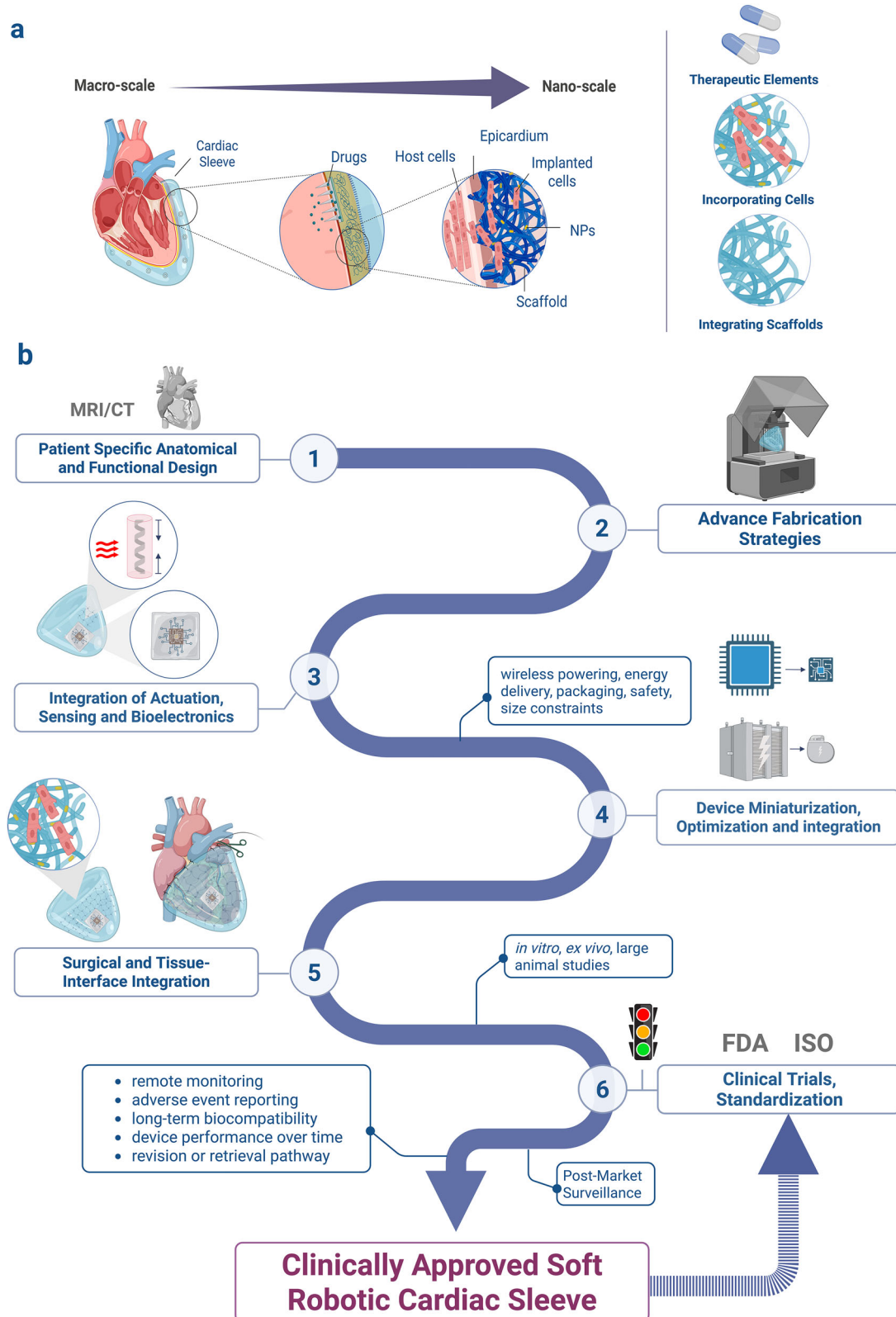
Expanding *in vitro* assays for soft robotic sleeves

As sleeve designs incorporate extracellular-matrix-mimetic hydrogels, bioadhesive layers, conductive polymers, anisotropic composites, or cellularised coatings, expanded assays are required to assess degradation behaviour, protein adsorption, oxidative stress, extracellular matrix remodelling, and long-term stability under cyclic loading^{90,91}. *In vitro* mechanobiology platforms enable systematic evaluation of how imposed compression, torsion, or strain modulate cardiomyocyte coupling, fibroblast activation, and epicardial cell phenotype, thereby identifying risks such as pro-fibrotic signalling or conduction disturbances before animal testing^{92,93}.

Electrophysiological *in vitro* studies are essential for evaluating the interaction between sleeve materials and cardiac electrical function. Culturing primary or induced pluripotent stem cells (iPSC)-derived cardiomyocytes on candidate biomaterials allows quantification of action potential morphology, conduction velocity, calcium handling, gap-junction expression, and cytoskeletal alignment in response to substrate stiffness, viscoelasticity, conductivity, and surface architecture^{78,94}.

Dynamic mechanical loading platforms that apply cyclic stretch, compression, or torsion within physiological ranges enable investigation of mechano-electrical feedback, stretch-activated ion channel responses, and load-dependent arrhythmogenic thresholds⁹⁵. Additional assessment is required for sleeves incorporating electrical or magnetic actuation, including pacing thresholds, capture reliability, and electromagnetic interference⁹⁶. Three-dimensional cardiac spheroids and organoids composed of cardiomyocytes, fibroblasts, endothelial cells, and epicardial-like populations provide higher-order models for studying synchronisation between intrinsic tissue contraction and external actuation⁹⁷.

Heart-on-a-chip and related microphysiological systems offer controlled environments that integrate engineered myocardium, microfluidics, and programmable mechanical actuation modules to recapitulate physiologically relevant cardiac microenvironments⁹⁸. These platforms enable continuous measurement of contractile force, electrophysiological stability, conduction velocity, metabolic activity, oxygen tension, and extracellular matrix remodelling^{98,99}. Co-culture configurations incorporating immune and stromal cells allow interrogation of inflammatory and fibrotic pathways, while integrated bio-mechanical stimulation systems can replicate sleeve-induced compression, torsional deformation, and cyclic mechanical loading patterns associated with soft robotic cardiac assist devices¹⁰⁰. In this context, heart-on-a-chip platforms may provide valuable intermediate preclinical systems for evaluating tissue-device interactions, mechano-responsive signalling, cardiomyocyte viability, fibrosis, and electro-mechanical synchronisation under physiologically relevant loading conditions. Furthermore, integration of biosensing modules and patient-specific tissue constructs may facilitate optimisation of actuator timing, compression magnitude, and control strategies prior to animal studies and clinical translation. Such systems can provide higher predictive value than conventional cytotoxicity assays and may enable earlier identification of subtle long-term biological and



biomechanical effects associated with soft robotic cardiac sleeve technologies¹⁰⁰.

Role of soft bioelectronics

Recent progress in soft bioelectronics has profoundly shaped the design space of cardiac-assist technologies, providing versatile

strategies that are directly translatable to soft robotic cardiac sleeves. The integration of stretchable conductive polymers, nanocomposite elastomers, and microfabricated sensor networks enables the creation of ultra-thin, conformal electronic systems capable of real-time electrophysiological, mechanical, and biochemical monitoring of the beating heart¹⁰¹. Using micro-transfer printing, inkjet lithography, thin-film

Fig. 6 | Bioengineering integration and translational roadmap strategies for soft robotic cardiac sleeves. **a** Conceptual overview illustrating multi-scale bioengineering approaches for enhancing soft robotic cardiac sleeves. At the macro scale, anatomically conformal sleeves provide mechanical cardiac assistance. At smaller scales, bioactive components can be incorporated to improve device-tissue integration and therapeutic functionality. These include scaffold-based architectures for structural support, incorporation of therapeutic elements such as drugs or conductive materials, and integration of cellular components to create biohybrid sleeve systems capable of interacting with the epicardial environment. Created with BioRender.com. **b** Translational roadmap for clinically viable soft robotic cardiac sleeves. Conceptual framework outlining key stages required for the clinical translation of soft robotic cardiac sleeves. The roadmap includes

patient-specific anatomical and functional design based on clinical imaging, advanced fabrication strategies, integration of actuation, sensing and bioelectronic components, and device miniaturisation with optimisation of power delivery and packaging. Subsequent stages involve surgical and tissue-interface integration, preclinical validation through in vitro, ex vivo and large-animal studies and progression to clinical trials and regulatory standardisation. Long-term post-market surveillance, including remote monitoring, adverse event reporting and assessment of device durability and biocompatibility, supports the pathway toward clinically approved soft robotic cardiac sleeve systems. Created with BioRender.com. Created in BioRender. Nazari, H. (<https://BioRender.com/supcva2>).

deposition, and additive three-dimensional (3D) manufacturing, these bioelectronic platforms achieve high mechanical compliance while preserving electrical fidelity under cyclic deformation¹⁰². Adapting these fabrication principles to cardiac sleeve architectures enables the co-integration of actuation layers, electrophysiological sensors, temperature or pH micro-detectors, and localised therapeutic delivery channels within a single anatomically adaptive construct, thereby extending sleeve functionality beyond mechanical assistance alone.

Such approaches are particularly relevant to soft robotic cardiac sleeves, where tether-free operation, continuous monitoring, and adaptive actuation are essential for long-term clinical deployment. However, the integration of wireless power transfer, signal stability, and long-term biocompatibility within mechanically dynamic cardiac environments remains incompletely resolved.

Beyond technical integration, the translation of soft bioelectronic cardiac systems requires alignment with regulatory and validation frameworks. The establishment of standardised pathways encompassing materials characterisation, electrical safety, preclinical evaluation, and regulatory compliance (including FDA, EMA, and ISO 10993 guidelines) is critical for ensuring safety, reproducibility, and scalability⁶³. engineered components, regulatory complexity may increase further, requiring consideration of combination product frameworks and additional medical electrical equipment standards, such as IEC 60601, alongside conventional biocompatibility testing.

Toward an ideal soft robotic cardiac sleeve: design principles and translational framework

Technical design considerations

The development of an ideal soft robotic cardiac sleeve requires the coordinated optimisation of biomedical, engineering, and clinical parameters within a patient-specific design paradigm¹⁰³. Accurate clinical diagnosis and comprehensive characterisation of cardiac pathology form the foundation of this approach, encompassing 3D cardiac geometry, spatial extent of myocardial injury, disease phenotype, patient age, and residual myocardial function. Adapting sleeve geometry and functional performance to anatomical and physiological variability enables personalised cardiac support that maximises mechanical efficiency while minimising invasiveness and adverse remodelling, establishing a pathway towards bioadaptive, regenerative, and clinically translatable cardiac assist systems. Importantly, the heterogeneous nature of HF, including ischaemic and non-ischaemic phenotypes, highlights the need for spatially resolved and patient-specific actuation strategies rather than uniform global compression. In ischaemic HF, regional akinesia or dyskinesia may require localised mechanical augmentation of infarct-adjacent myocardium, whereas diffuse cardiomyopathies may benefit from broader ventricular support profiles. Future sleeve architectures require independently addressable actuator arrays capable of modulating force, timing, and deformation patterns across different ventricular regions.

At the core of soft robotic cardiac sleeve function lies the contraction mechanism, which governs the influence on ventricular loading and synchrony. Early-generation sleeves predominantly relied

on pneumatic or hydraulic actuation, achieving physiologically relevant stroke volumes but remaining constrained by external compressors, high power demands, and limited implantability^{61,64}. Although these systems demonstrated robust proof-of-concept, factors such as performance, tethered operation and limited actuation bandwidth restricted their suitability for chronic deployment. Current development trajectories increasingly favour electrothermal, EAP, and hybrid actuation strategies⁶⁷. Electrothermal actuators employ Joule heating of conductive filaments (for example, silver-coated nylon, carbon nanotube fibres, or graphene-based conductors) embedded within elastomeric matrices, enabling programmable deformation without external pressure lines¹⁴. Dielectric EAPs provide voltage-controlled strain with rapid response and minimal acoustic noise, aligning with the rhythmic demands of cardiac pacing⁴³. Integration of anisotropic reinforcement and controlled fibre orientation is essential to reproduce native myocardial torsion and strain distribution¹⁰⁴. Stimuli-responsive hydrogels¹⁰⁵, shape-memory alloys¹⁰⁶, and liquid metal composites¹⁰⁷ can further enhance force output and fatigue resistance. Recent studies on left and right atrioventricular coupling emphasise the importance of ventricular filling dynamics, chamber compliance, pressure-volume relationships, and electromechanical synchrony in maintaining coordinated cardiac performance¹⁰⁸. Consequently, advanced soft robotic sleeves may benefit from architectures capable of preserving septal motion, interventricular coupling, and pulmonary haemodynamics while dynamically adjusting compression profiles during disease progression or recovery.

Manufacturing strategy directly constrains both structural fidelity and functional complexity³⁹. Building on early demonstrations of fully conformal integumentary electronic membranes, contemporary fabrication approaches integrate hybrid additive manufacturing, micro-channel embedding, and bioactive hydrogel patterning to enable multifunctional architectures¹⁰⁹. Advanced processes such as direct ink writing (DIW), conductive ink embedding, and multi-material printing allow simultaneous fabrication of actuators, sensors, and interconnects, supporting monolithic, functionally graded constructs rather than mechanically assembled components^{110,111}. Recent advances in micro- and nanofabrication, including electrospinning, textile-based reinforcement, and decellularised tissue processing, have enabled the development of layered and fibre-aligned architectures capable of reproducing myocardial anisotropy and directional force transmission⁹⁰.

Although patient-specific cardiac sleeve architectures may improve anatomical conformity and biomechanical integration, such approaches may also introduce important challenges related to manufacturing scalability, procedural standardisation, and healthcare cost. The requirement for personalised imaging, custom fabrication, integrated bioelectronics, and specialised surgical implantation may initially limit widespread accessibility compared with established heart failure therapies such as durable VADs or transplantation. Clinical translation will therefore likely depend on advances in scalable manufacturing workflows, modular device architectures, automated fabrication strategies, and streamlined regulatory pathways capable of balancing patient-specific optimisation with practical clinical implementation.

Device intelligence arises from the closed-loop integration of actuation and sensing systems¹¹². Soft bioelectronics enable continuous electrophysiological and haemodynamic monitoring without compromising mechanical compliance. Elastomer-embedded nanocomposites provide stretchable, low-impedance electrodes and distributed sensing arrays¹¹³. In feedback-controlled architectures, embedded sensors detect myocardial strain, ECG signals, or pressure fluctuations, while onboard controllers dynamically adjust actuation timing and amplitude^{60,113}. The incorporation of therapeutic functions (such as local drug or extracellular vesicle (EV) release triggered by electrical, thermal, or biochemical cues) extends the sleeve beyond mechanical support towards an intelligent therapeutic biointerface. Future systems need to incorporate multimodal closed-loop control integrating electrophysiological, haemodynamic, and biomechanical feedback to enable adaptive and autonomous cardiac assistance.

Robust integration of structural, electronic, and biological components is essential for long-term performance and safety. Uniform stress distribution across layered architectures is required to prevent delamination and focal pressure injury¹¹⁴. Interpenetrating polymer networks, gradient interfaces, and self-healing elastomers may improve fatigue resistance under millions of cardiac cycles¹¹⁵. At the tissue interface, conductive hydrogels or decellularised extracellular matrix (ECM) coatings can mitigate immune activation and enhance electrical coupling^{78,116}. Importantly, epicardial adhesion must preserve coronary perfusion, and mechanical compliance between actuation and sensing layers must be maintained to ensure signal fidelity under repeated deformation¹⁰¹.

Translational and clinical framework

Beyond device design, surgical implantation introduces several translational challenges. Effective clinical adoption requires standardised, reproducible implantation workflows that preserve pericardial integrity, avoid coronary compression, and protect adjacent structures such as the phrenic nerve and atrioventricular groove. While most existing studies employ median sternotomy with partial pericardiotomy, minimally invasive and robot-assisted approaches may become feasible as device profiles and fixation strategies evolve. Achieving uniform epicardial coverage without excessive manipulation or suturing is critical, as epicardial trauma increases the risk of bleeding and postoperative arrhythmias. Clear decision pathways for device explantation or revision are particularly important for biointegrative or adhesive-based designs, where long-term attachment may complicate retrieval. Chronic implantation raises concerns regarding fibrotic encapsulation and persistent inflammatory responses, phenomena observed across long-term cardiac implants and requiring systematic evaluation. One of the most important translational considerations involves the interaction between sleeve-assisted biomechanics and cardiac reverse remodelling during chronic support. Mechanical unloading and externally applied ventricular assistance may progressively alter ventricular geometry, wall stress, chamber compliance, and myocardial strain distribution over time^{117,118}. Consequently, future cardiac sleeves may require adaptive or reconfigurable actuation systems capable of responding to evolving cardiac anatomy and functional recovery during long-term implantation.

Appropriate patient selection will also represent a critical determinant of future clinical success¹¹⁹. Soft robotic cardiac sleeves may be particularly suited for patients with advanced systolic heart failure who retain partially recoverable myocardial function and could benefit from mechanical augmentation without direct blood contact. In contrast, patients with severe myocardial fibrosis, restrictive physiology, advanced right ventricular failure, or extensive electrical dyssynchrony may present additional biomechanical and surgical challenges.

Miniaturisation and mass optimisation are central to chronic deployment. Sleeve architectures must combine anatomical conformity with low mass and energy demand, favouring materials with

tissue-like density ($< 1 \text{ g cm}^{-3}$) and low elastic modulus (10–100 kPa) to minimise mechanical mismatch⁵⁰. Patient-specific finite element modelling informed by MRI or CT imaging enables tailoring of geometry, actuation strength, and thermal load. Energy-efficient actuation, compact stretchable electronics, and low-profile sensors are essential to ensure reliable long-term operation without overheating or overcompression, while modular connectivity may simplify surgical handling and maintenance^{50,120}. Wireless power transfer, implantable energy storage systems, and low-profile control electronics will likely become increasingly important for fully implantable chronic cardiac support platforms³².

Despite advances in implantable bioelectronics and wireless energy transfer, the long-term clinical feasibility of fully implantable soft robotic cardiac assist systems remains uncertain. Major translational challenges include chronic device reliability, infection prevention, battery longevity, thermal safety, surgical reproducibility, and stable long-term electromechanical integration. Importantly, fully wireless ventricular assist or mechanical circulatory support systems have not yet achieved widespread clinical implementation, highlighting the extensive engineering and regulatory barriers that remain for chronically implantable cardiac sleeve platforms.

These considerations highlight that successful translation of non-blood-contact cardiac assist technologies extends beyond device mechanics, requiring careful alignment between patient phenotype, therapeutic intent, and timing of intervention. In parallel, heart-on-a-chip and related microphysiological systems may support translational development by enabling controlled evaluation of sleeve-induced mechanical deformation, tissue remodelling, electrophysiological responses, and inflammatory signalling under physiologically relevant conditions. Such systems may provide higher predictive value than conventional cytotoxicity assays while reducing reliance on early-stage animal experimentation.

Figure 6b proposes an integrated translational framework that brings together the multidisciplinary engineering, biological, manufacturing, regulatory, and clinical considerations required to advance soft robotic cardiac sleeves from conceptual prototypes towards clinically viable therapeutic platforms. The framework highlights the sequential and interdependent milestones spanning patient-specific anatomical design, advanced manufacturing, bioelectronic integration, device optimisation, preclinical validation, regulatory translation, and long-term clinical surveillance, providing a holistic roadmap for future development and implementation.

The major engineering, biological, and translational priorities likely to shape the next generation of clinically viable soft robotic cardiac sleeves are summarised in Box 2.

Conclusion

Soft robotic cardiac sleeves represent an emerging class of cardiac assist technologies that provide physiologically relevant, non-blood-contact ventricular support. By mimicking native myocardial motion while avoiding direct interaction with circulating blood, these systems have the potential to reduce thrombogenic and infectious risks associated with intravascular mechanical assist devices.

Despite these advantages, translation into long-term animal studies and routine clinical practice remains constrained by several unresolved challenges. Although acute and short-term preclinical investigations consistently demonstrate haemodynamic benefit, durable epicardial integration under sustained mechanical actuation remains a principal barrier to chronic implantation. Achieving effective force transmission, biological tolerance, coronary safety and device reversibility simultaneously has not yet been realised with existing interface materials and fixation strategies. Additional limitations include ensuring long-term biocompatibility under cyclic loading, developing implantable and energy-efficient power and control systems, and validating sustained mechanical performance in models of

BOX 2

Key priorities for next-generation soft robotic cardiac sleeves

- Regionally targeted and patient-specific actuation strategies
- Adaptive assistance for ischaemic versus non-ischaemic HF
- Closed-loop control integrating haemodynamic and electromechanical feedback
- Long-term epicardial biointegration and reduction of inflammatory encapsulation
- Miniaturised implantable power delivery and wireless control systems
- Chronic large-animal validation and reverse-remodelling assessment
- Standardised benchmarking protocols for device comparison
- Integration of heart-on-a-chip and biohybrid cardiovascular models for preclinical testing
- Scalable manufacturing, sterilisation compatibility, and regulatory readiness
- Long-term post-market surveillance and retrievability considerations.

chronic cardiac disease. In parallel, the absence of clearly defined regulatory pathways and standardised surgical workflows continues to impede clinical deployment, particularly for multifunctional and biointegrative systems.

Successful clinical translation of soft robotic cardiac sleeves will therefore depend on tightly coordinated, multidisciplinary development integrating cardiology, cardiac surgery, biomaterials science, soft robotics and bioelectronics engineering. Within such collaborative frameworks, soft robotic sleeves may evolve into patient-specific therapeutic platforms capable of addressing the heterogeneity of cardiac pathology, thereby expanding the clinically viable treatment options for patients with advanced HF.

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Author contributions

J.F. conceived the Review, supervised the project, and developed the overall framework. H.N. conducted the literature review, prepared the figures, and wrote the manuscript. J.F. and H.N. interpreted the literature and refined the scientific scope of the Review. N.L., C.H., C.H.W., and A.R. discussed, critically revised, and edited the manuscript. All authors approved the final manuscript.

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